

IoT Based Automated Table Lamp

Rohan Barnwal

This is an IoT based table lamp that works in such a way that whenever a person sits on the chair it automatically turns on and when the person leaves the chair it automatically turns off. It also has bed-time alert feature to indicate it's time to sleep when it turns off the lamp automatically and gives 'Time to Sleep' notification on your mobile phone. You can also manually control

the lamp on/off from the Blynk app in your smartphone.

Circuit and working

At the heart of this project are two NodeMCU ESP8266 boards, Board1 and Board2, and an IR sensor module. Board1 with IR sensor is used as transmitter to send the signals collected by the sensor, which is fixed on the chair, through a mobile network. Board2

with the lamp is used as receiver to collect the signals.

The IR sensor connected to Board1 checks whether a person is sitting on the chair or not. Whenever a person sits on the chair the state of IR sensor becomes high. The IR sensor always sends real-time data to the receiver.

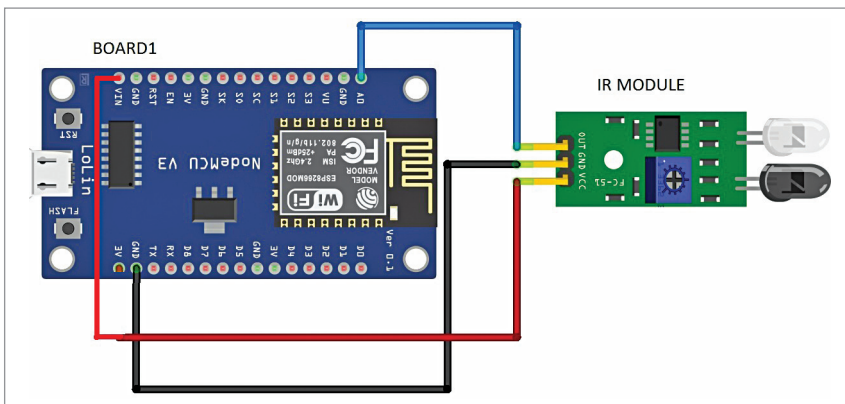


Fig. 1: Connection diagram of IR transmitter

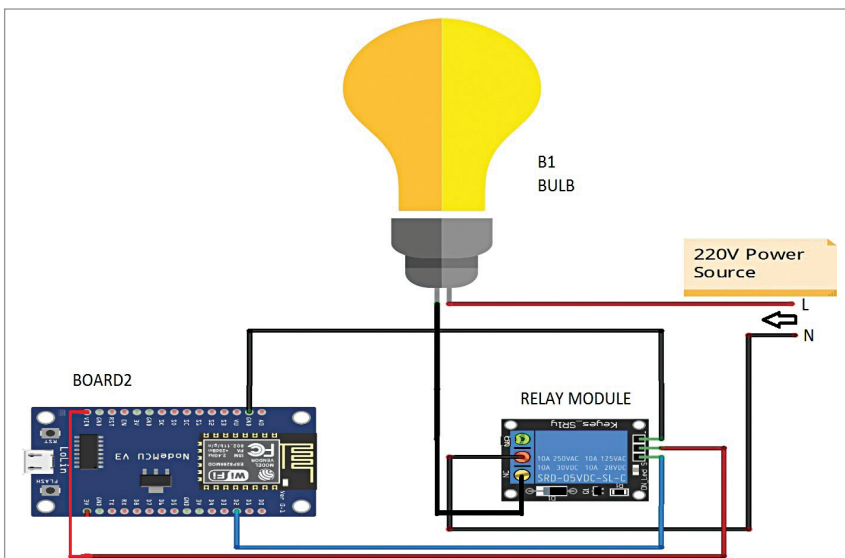


Fig. 2: Connection diagram of receiver

Connections of IR Sensor to NodeMCU		Table 1
IR sensor module pin	NodeMCU (Board1) pin	
GND	G	
Vcc	VIN	
OUT	AO	

Connections of Relay Module to NodeMCU		Table 2
Relay module pin	NodeMCU (Board2) pin	
Vcc	VIN	
GND	G	
OUT	D2	

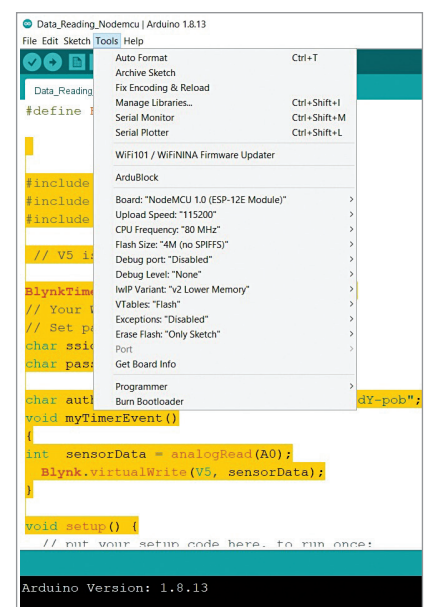


Fig. 3: Board selection